



Archit Sharma

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Gender: Male Date of birth: 13/01/2001 Nationality: Indian



EDUCATION & TRAINING

[01/10/2020 - 31/05/2024]

Bachelors of Technology in Computer Science and Engineering

DIT University

- B.Tech in Computer Science & Engineering — Specialization: AI, Machine Learning & Robotics
- Machine learning theory and practice: supervised, unsupervised, and reinforcement learning
- Deep learning architectures: CNNs, RNNs, Transformers, and generative models
- Fuzzy logic systems and approximate reasoning
- Evolutionary computing and nature-inspired optimization: genetic algorithms, swarm intelligence, evolutionary strategies
- Computer vision fundamentals and applications
- Autonomous systems design, sensor fusion, and ROS-based robotics
- Simulation environments for robotics development
- Data structures, algorithms, and object-oriented design
- Software engineering principles and applied programming in Python

[01/10/2025 - Current]

Master of Science in Artificial Intelligence and Machine Learning

Technische Universität Darmstadt

- M.Sc. in Artificial Intelligence / Machine Learning
- Probabilistic graphical models, Bayesian networks, and inference algorithms
- Causal discovery and causal inference: do-calculus, PC, FCI, and GES algorithms
- Causal representation learning and multimodal AI systems
- Deep learning theory and architecture design: vision-language models, diffusion models, CLIP, DALL-E
- Reinforcement learning and generative modelling
- Multimodal AI for robotic learning: vision-language-action models, embodied AI, perception-action integration, and sim-to-real transfer
- Ethics in NLP: fairness, bias mitigation, transparency, and responsible LLM deployment
- Research methodology: experimental design, academic writing, and independent research
- Practical project experience in causal reasoning, visual perception, and graph-based AI tools

WORK EXPERIENCE

EffiGO Global

City: Hyderabad | Country: India

[03/06/2024 - 16/07/2025]

Product Engineer

- Designed and developed AI/ML algorithms for NLP-based document generation and intelligent recommendation systems
- Deployed scalable ML solutions on Google Cloud Vertex AI, Amazon Bedrock, and AWS SageMaker across GCP, AWS, and Azure environments
- Built generative AI applications using Gemini and Titan models for automated RFQ generation and contract automation
- Architected end-to-end AI pipelines with serverless architectures and containerized microservices
- Managed full MLOps lifecycle: model versioning, monitoring, continuous training pipelines, and automated deployment workflows
- Collaborated on React and Spring-based frontends with integrated AI-powered chatbot and conversational interfaces

EffiGO Global

City: Bangaluru | Country: India

[01/02/2024 - 31/05/2024]

Product Engineer Intern

- Trained in Spring Boot application development and RESTful API design principles
- Developed a chatbot application for automated query resolution and end-user support
- Engineered an OCR-based solution for automated invoice processing and purchase order matching

PROJECTS

[10/11/2025 - 15/04/2026]

Causeway

- Designed and developed an open-source causal AI toolkit for visualizing causal structures, inferring causal relationships from CSV data, and extracting causal graphs from images
- Implemented causal discovery algorithms and interactive graph-based representations using probabilistic graphical model principles
- Built end-to-end pipeline integrating data ingestion, causal inference, and visual output for exploratory causal analysis

Link: <https://github.com/DEMONKINGKAI/Causeway>

[20/05/2026 - Current]

Threadfall — The Shattered Pact

- Designed and built a solo narrative RPG engine governed entirely by causal inference rather than randomness, implementing Judea Pearl's full three-rung Ladder of Causation (association, intervention, counterfactual)
- Engineered a structural causal DAG (20 nodes, 26 edges, 5 acts) with do-calculus graph surgery and topological propagation for deterministic, reproducible story outcomes
- Integrated a discrete Bayesian Network (pgmpy) with auto-generated CPTs and Variable Elimination inference for real-time probabilistic world-state belief updates
- Developed a verb-first intent classifier mapping free-text player input to canonical action types without LLM dependency, ensuring low-latency and interpretable classification
- Architected a full-stack application (FastAPI backend, React frontend) with LLM narration layer (Qwen 2.5) strictly constrained to prose generation — causal engine retains full decision authority

Link: <https://github.com/DEMONKINGKAI/Threadfall>

CERTIFICATIONS

[Microsoft, 24/08/2025]

Microsoft AI Product Manager

[Politecnico di Milano]

Ethics of Artificial Intelligence

[University of Michigan]

Applied Machine Learning in Python

SKILLS

Machine Learning | Deep Learning | Reinforcement Learning | Evolutionary Computing | NLP | Large Language Models (LLMs) | Diffusion Models | Multimodal AI | MLOps | PyTorch | Google Cloud Platform (GCP) | Amazon Web Services (AWS) | OpenCV | Microsoft Azure | Containerization (Docker/Kubernetes) | Python | Java | Backend Development | Computer Vision | Vision-Language Models | Git | SQL | optical character recognition software | TensorFlow | LangChain | Llama Index

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

English

LISTENING: C2 READING: C1 WRITING: C1

SPOKEN PRODUCTION: C1 SPOKEN INTERACTION: C1